

Eagle Scout Project

Renovation of Great Escape Early Learning Center Playgrounds

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Project Overview

Parameter	Detail
Project Title	Renovation of Two Front Playgrounds
Location	Great Escape Early Learning Center, Littleton, Colorado
Sponsor	Littleton United Methodist Church & Troop 114
Advisor	John Grace (Retired Systems Engineering Manager, Space Systems Company)
Duration (Personal)	40 Hours Pre-Planning & 30 Hours Execution
Volunteers	40+
Total Service Hours	≥ 310 man-hours

Table 1: Eagle Scout Project Summary

Led a team of over 40 volunteers in the complete renovation of two front playgrounds at the Great Escape Early Learning Center. The project significantly improved safety, accessibility, and structural integrity by deepening the play areas, replacing deteriorated containment borders, refreshing the safety surfacing, and restoring existing play equipment.



Figure 1: Before renovation — one of the two playground areas

Project Planning and Coordination

Coordinated all aspects of the project with church leadership and early learning center staff. This included:

- Developing a detailed project schedule and timeline
- Creating a project budget and securing necessary funding/materials
- Obtaining required permissions and coordinating site access
- Minimizing operational disruptions to daily childcare activities

Regular communication with the sponsor (Littleton United Methodist Church) and the Early Learning Center ensured the project aligned with their operational needs and safety standards.

Site Preparation and Earthwork

- Excavated and regraded both playground surfaces by removing the top layer of compacted gravel.
- Increased the depth of the play areas to accommodate additional impact-absorbing surfacing material.
- Established proper drainage slopes and created a level, stable sub-base.



Figure 2: Excavation and regrading of playground surfaces

Playground Border Replacement

Demolished the deteriorated wooden playground borders on both play areas and installed a new, robust containment system.

- Installed new pressure-treated landscape timbers, vertically anchored with steel rebar stakes driven into the ground for long-term stability.
- Capped the timbers with heavy-duty plastic playground edging barriers to securely contain the safety surfacing and improve edge safety.



Figure 3: Installation of new rebar-anchored timber borders with plastic edging

Safety Surfacing and Equipment Restoration

- Screened (sieved) the existing gravel to remove debris, sharp objects, and foreign material.
- Supplemented the processed gravel with additional clean material and leveled it to achieve the required depth for fall protection compliance.
- Sanded down and refinished all existing play structures to eliminate splinters, rough edges, and weathered surfaces, significantly improving equipment safety and longevity.



(a) Playground 1



(b) Playground 2

Figure 4: Completed playground after surfacing and equipment restoration

Volunteer Leadership and Management

Recruited, organized, and led a volunteer crew of over 40 individuals, including Boy Scout troop members, friends, and church congregants. The project was completed over three work days with a cumulative total exceeding 310 man-hours.

Key leadership responsibilities included:

- Task delegation and workflow optimization
- Safety briefings and enforcement of safe work practices
- Material procurement and on-site logistics
- Quality control and final site cleanup

Project Impact

The renovated playgrounds now provide a significantly safer, cleaner, and more enjoyable environment for the young children at the Great Escape Early Learning Center. The deepened safety surfacing, reinforced borders, and restored equipment substantially reduce injury risk while extending the usable life of the playground facilities.

Skills Demonstrated

Category	Skills Demonstrated
Project Management	Scheduling, budgeting, stakeholder coordination, permitting
Leadership	Recruiting and leading 40+ volunteers, task delegation, safety management
Construction	Earthwork, structural border installation, material screening and grading
Communication	Coordination with church, learning center, and adult advisors
Documentation	Project planning, material lists, before/after records

Table 2: Key Skills Demonstrated During Eagle Scout Project

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